Contraction, Atomic Capacity, and Exact One-Seed Simulation

Abstract

We separate contraction of unobserved states from the entropy and atomic structure of observed trajectories. Every finite controlled classical law can be realized by instruments with the same prescribed nonselective channel. After fixing a binary measurement instrument, we obtain exact formulas for finite-horizon seed entropy and a complete classification of atomic support and asymptotic forgetting. Complete forgetting may coexist with zero entropy rate and the impossibility of a countably supported infinite-horizon seed. Conversely, almost-sure stabilization may coexist with infinite seed entropy. All statements below are proved directly; no claim of historical priority is made.

1 Seeds and atomic mass

All Shannon entropies are in bits; ln denotes the natural logarithm. Set $h_2(p) = -p\log_2 p - (1-p)\log_2(1-p)$, with $0\log_2 0 = 0$. Let $\mathcal{Y}$ be finite and μ a probability measure on $\Omega = \mathcal{Y}^{\mathbb{N}_0}$ with its product sigma algebra. Write $Y_{0:n} = (Y_0, \dots, Y_n)$ and $\mu_n = \mathcal{L}(Y_{0:n})$.

Definition 1.1. The finite-horizon seed cost is

$$C_n = \inf\{H(S) : S \text{ is countably valued, } F(S) \sim \mu_n\}.$$

The infinite-horizon cost C_∞ is defined similarly with a measurable map F into Ω. The infimum of the empty set is $+\infty$. The atomic mass is

$$\mathrm{at}(\mu) = \sum_{\omega:\, \mu\{\omega\}>0} \mu\{\omega\}.$$

For this one-action setting, define atomic capacity as the supremum of the total point-atom mass of a seed law on a standard Borel space that admits a measurable deterministic decoder with output law μ.

Proposition 1.2. *One has $C_n = H(Y_{0:n})$. Atomic capacity equals $\mathrm{at}(\mu)$. A countably supported exact infinite-horizon seed exists if and only if $\mathrm{at}(\mu) = 1$. If this holds, the trajectory itself is an optimal seed and*

$$C_\infty = H(\mu) = \sup_n H(Y_{0:n}),$$

including the value $+\infty$. In all cases $C_\infty = \sup_n C_n$.

Proof. For finite horizons, deterministic data processing gives $H(Y_{0:n}) \le H(S)$, and $S = Y_{0:n}$ attains equality. Every seed atom maps to an output atom. The total mass of seed atoms is therefore at most $\mathrm{at}(\mu)$; the identity decoder on (Ω, μ) attains this bound. A countable seed has countable image, and conversely a purely atomic trajectory law can itself be used as a countable seed.

Let $[\omega]_{0:n}$ be the cylinder determined by the first $n+1$ coordinates. Then

$$I_n(\omega) = -\log_2 \mu([\omega]_{0:n}) \uparrow -\log_2 \mu\{\omega\},$$

because the cylinders decrease to $\{\omega\}$. Zero-mass cylinders form a null set and may be omitted. Monotone convergence proves $\sup_n H(\mu_n) = \int -\log_2 \mu\{\omega\}\, d\mu(\omega)$. If μ is purely atomic, this is its discrete entropy. Deterministic data processing gives the optimality assertion. If $\mathrm{at}(\mu) < 1$, the integrand is infinite on a positive-measure set, so the supremum is infinite, as is C_∞. □

2 Channels do not determine instrument laws

Let $\mathcal{H}$ be a finite-dimensional Hilbert space. A channel is a completely positive trace-preserving linear map on matrices. A finite instrument is a family of completely positive trace-nonincreasing maps whose sum is a channel. Its outcome probability in state ρ is $\operatorname{Tr} \mathcal{I}_y(\rho)$.

Theorem 2.1 (Instrument non-identifiability)**.** *Fix a channel T. For every finite controlled tree with nonempty finite local alphabets and transition laws $P(y \mid h, a)$, there are instruments at its histories satisfying*

$$\sum_y \mathcal{I}_{h,a,y} = T$$

whose output law agrees with that tree under every deterministic adaptive policy. The same assertion holds for a countable sequence of finite levels.

Proof. Define, for every matrix X,

$$\mathcal{I}_{h,a,y}(X) = P(y \mid h, a) T(X).$$

Scalar multiplication by a nonnegative number preserves complete positivity. On positive matrices the trace is multiplied by a number in $[0, 1]$, and summing over y gives T. For a normalized input state, $\operatorname{Tr} \mathcal{I}_{h,a,y}(\rho) = P(y \mid h, a)$. Conditional on a positive probability outcome, the normalized next state is $T(\rho)$. Induction on history length proves equality of all cylinder laws under every policy. The instruments at a shared history are the same, independently of which policy reached it. At infinite horizon equality of all cylinders determines the trajectory law. □

Corollary 2.2. *The nonselective channel, even when specified exactly, does not determine finite-horizon causal seed cost, infinite-horizon atomic capacity, or output entropy rate. This holds even for stationary one-action instruments: $\mathcal{I}_0 = T$ has deterministic output, while $\mathcal{I}_0 = \mathcal{I}_1 = T/2$ has independent fair outputs.*

Remark 2.3. Theorem 2.1 permits history-dependent instruments and unrestricted instrument randomness. It is not an assertion under a fixed measurement, a fixed dilation, a closed-system model, or a bound on external memory or randomness. Those are additional restrictions. For a controlled tree the one-seed requirement concerns all policies; it is not the entropy of one arbitrarily chosen policy transcript.

3 Exact contraction and its limits

For a density matrix σ, a unitary U, and $0 \le q \le 1$, set

$$T(X) = qUXU^\dagger + (1 - q) \operatorname{Tr}(X)\sigma.$$

The trace factor makes the formula a linear map on all matrices.

Proposition 3.1. *For density matrices ρ, η,*

$$\|T(\rho) - T(\eta)\|_1 = q\|\rho - \eta\|_1.$$

For a prescribed sequence of such channels with parameters q_j, the distance after n steps is exactly the initial distance times $\prod_{j=1}^{n} q_j$. If all $q_j > 0$ and the initial distance is positive, it converges to zero if and only if $\sum_j -\ln q_j = \infty$. A zero factor erases the difference at that step.

Proof. The reset terms cancel on differences of trace-one matrices. Unitary conjugation preserves trace norm. Iterate and take logarithms of the product. □

Remark 3.2. The two compared states must undergo the same prescribed channel sequence. Feedback that chooses different channels from the states or from different observed histories does not satisfy this hypothesis. Changing U alone leaves the contraction factor unchanged.

4 A fixed measured reset process

Let $P_y = |y\rangle\langle y|$ for $y \in \{0, 1\}$ and $\mathcal{D}_Z(X) = P_0 X P_0 + P_1 X P_1$. Start in $I/2$ and measure Z to obtain Y_0. For each $j \geq 1$, choose $0 < q_j \leq 1$ and use the instrument

$$T_j(X) = q_j X + (1 - q_j)\,\mathrm{Tr}(X) I/2, \qquad \mathcal{I}_{j,y}(X) = P_y T_j(X) P_y.$$

Its nonselective channel is $\Phi_j = \mathcal{D}_Z \circ T_j$, not T_j. Put $p_j = (1 - q_j)/2 \in [0, 1/2)$.

Theorem 4.1 (Complete classification)**.** *The output process admits the representation*

$$Y_j = Y_{j-1} \oplus B_j, \quad Y_0 \sim \text{Bernoulli}(1/2), \quad B_j \sim \text{Bernoulli}(p_j),$$

where $Y_0, B_1, B_2, \ldots$ are independent. Consequently:

$$C_n = 1 + \sum_{j=1}^{n} h_2(p_j), \tag{1}$$

$$\mathrm{at}(\mu) = \begin{cases} 1, & \sum_j p_j < \infty, \\ 0, & \sum_j p_j = \infty, \end{cases} \tag{2}$$

$$C_\infty < \infty \iff \sum_j h_2(p_j) < \infty. \tag{3}$$

For distinct diagonal initial states ρ, η, their unobserved distance after n measured steps is

$$D_n = \|\Phi_n \cdots \Phi_1(\rho) - \Phi_n \cdots \Phi_1(\eta)\|_1 = D_0 \prod_{j=1}^{n} q_j.$$

In particular,

$$\boxed{D_n \longrightarrow 0 \iff \sum_j p_j = \infty \iff \mathrm{at}(\mu) = 0.}$$

The distance compares unobserved evolutions; it is not a distance between posterior states conditioned on a common or different record.

Proof. After outcome y, the state is P_y. The next outcome is $1-y$ with probability $(1-q_j)/2 = p_j$, independently of earlier history conditional on y. The bijection between output prefixes and $(Y_0, B_1, \dots, B_n)$ gives the asserted independent representation and the entropy formula, using Proposition 1.2.

If $\sum_j p_j < \infty$, the first Borel–Cantelli lemma gives only finitely many flips almost surely. There are countably many finite subsets of $\mathbb{N}$, hence countably many such records, so the law is purely atomic. If $\sum_j p_j = \infty$, every prefix has mass at most

$$\frac{1}{2}\prod_{j=1}^{n}(1-p_j) \le \frac{1}{2}\exp\left(-\sum_{j=1}^{n} p_j\right) \longrightarrow 0.$$

Thus each infinite record has mass zero. Proposition 1.2 now proves (3) as well.

On diagonal trace-zero matrices Φ_j acts by scalar multiplication by q_j. This proves the distance formula. Since $-\ln(1-2p) \ge 2p$, divergence of $\sum_j p_j$ forces the product to vanish. If the sum converges, $p_j \to 0$, and eventually $p_j \le 1/4$, so $-\ln(1-2p_j) \le 4p_j$. The initial finite product is positive because $q_j > 0$. Thus the infinite product is positive. □

Remark 4.2 (Zero factors and off-diagonal states)**.** If finitely many q_j vanish, complete forgetting can occur even when $\sum_j p_j < \infty$. Positivity of every q_j is essential for the displayed equivalence. Also, $\mathcal{D}_Z$ erases off-diagonal differences immediately; the exact distance assertion is restricted to diagonal states.

Corollary 4.3 (Three regimes)**.** *The following are the only possibilities in Theorem 4.1:*

Condition	$D_n \to 0$	*Countable seed*	*Finite-entropy seed*
$\sum h_2(p_j) < \infty$	*No*	*Yes*	*Yes*
$\sum p_j < \infty$, $\sum h_2(p_j) = \infty$	*No*	*Yes*	*No*
$\sum p_j = \infty$	*Yes*	*No*	*No*

Proof. For $0 \le p \le 1/2$, concavity of binary entropy gives $h_2(p) \ge 2p$. Combine this with Theorem 4.1. □

5 Sharp examples

Example 5.1 (Complete forgetting with zero entropy rate)**.** Take $p_j = 1/(j+2)$, so $q_j = j/(j+2)$. Then

$$\frac{D_n}{D_0} = \frac{2}{(n+1)(n+2)}, \qquad \mathrm{at}(\mu) = 0,$$

whereas

$$C_n = \frac{(\ln n)^2}{2\ln 2} + O(\ln n), \qquad C_n/n \to 0.$$

Indeed, the distance product telescopes. For $p \to 0$,

$$h_2(p) = \frac{p\ln(1/p) + p + O(p^2)}{\ln 2}.$$

The sum of $\ln(j+2)/(j+2)$ equals $\frac{1}{2}(\ln n)^2 + O(1)$, while $\sum_{j\le n} 1/(j+2) = O(\ln n)$ and $\sum_j 1/(j+2)^2 < \infty$.

Example 5.2 (Eventual stabilization with infinite seed entropy)**.** Take $p_j = 1/((j+2)(\ln(j+2))^2)$. These probabilities lie in $(0, 1/2)$. Then $\sum_j p_j < \infty$, so Y_j is eventually constant almost surely, $\mathrm{at}(\mu) = 1$, and D_n has a positive limit. Nevertheless

$$C_n = \frac{\ln \ln n}{\ln 2} + O(1) \longrightarrow \infty.$$

To verify this, write $x = j + 2$ and expand

$$p_j \ln(1/p_j) = \frac{1}{x \ln x} + \frac{2 \ln \ln x}{x (\ln x)^2}.$$

The second term is summable, as are p_j and p_j^2; the sum of the first term is $\ln \ln n + O(1)$. Proposition 1.2 gives $C_\infty = \infty$ despite countable support.

Example 5.3 (Finite entropy)**.** For $p_j = 1/(j+2)^2$, both $\sum p_j$ and $\sum h_2(p_j)$ converge. Thus a single countable seed with finite entropy realizes every horizon.

6 No universal divergent lower scale

Theorem 6.1. *For every nondecreasing function $g : \mathbb{N} \to (0, \infty)$ with $g(n) \to \infty$, there exists a sequence $0 < q_j \leq 1$ in Theorem 4.1 such that*

$$D_n \to 0, \qquad \mathrm{at}(\mu) = 0, \qquad C_n \to \infty, \qquad C_n = o(g(n)).$$

Moreover, the sequence can be chosen with $p_j > 0$ at every time.

Proof. Choose strictly increasing integers n_k such that $g(n_k) \geq k^2$. Put $p_{n_k} = 1/4$ and initially put $p_j = 0$ otherwise. If $N(n) = \#\{k : n_k \leq n\}$, then

$$C_n = 1 + h_2(1/4) N(n), \qquad D_n / D_0 = 2^{-N(n)}.$$

For $n_k \leq n < n_{k+1}$, the ratio $C_n / g(n)$ is at most $(1 + h_2(1/4)k)/k^2$, which tends to zero. Infinitely many designated times give $\sum p_j = \infty$. To obtain strict positivity at every time, replace the remaining zero probabilities by 2^{-j-3}. Their total binary entropy is finite, because $h_2(p) \leq [p \ln(1/p) + p]/\ln 2$ for $0 < p < 1$. This changes C_n by a uniformly bounded amount and preserves every conclusion. □

7 A uniform perturbation estimate

The product identity requires identical channels. The following estimate handles different dynamics.

Proposition 7.1. *Let T_j be trace-distance contractions on states with factors q_j, and let $\widetilde{T}_j$ be channels satisfying*

$$\sup_\rho \|T_j(\rho) - \widetilde{T}_j(\rho)\|_1 \leq \varepsilon_j.$$

If $\rho_j = T_j(\rho_{j-1})$ and $\widetilde{\rho}_j = \widetilde{T}_j(\widetilde{\rho}_{j-1})$, then

$$\|\rho_n - \widetilde{\rho}_n\|_1 \leq \left(\prod_{j=1}^n q_j \right) \|\rho_0 - \widetilde{\rho}_0\|_1 + \sum_{k=1}^n \varepsilon_k \prod_{j=k+1}^n q_j.$$

If $q_j \leq q < 1$ and $\varepsilon_j \leq \varepsilon$, the right side is at most $q^n \|\rho_0 - \widetilde{\rho}_0\|_1 + \varepsilon(1 - q^n)/(1 - q)$.

Proof. The triangle inequality gives $d_j \le q_j d_{j-1} + \varepsilon_j$. Iterate this scalar inequality and sum the geometric series. □

Remark 7.2. This estimate concerns the unobserved state only. It does not bound the total variation distance of entire records when the instruments differ; Theorem 2.1 supplies counterexamples even when $\varepsilon_j = 0$ for every j.

8 Finite data cannot identify infinite atomic capacity

Theorem 8.1. *Fix any finite prefix $p_1, \dots, p_N \in [0, 1/2)$. It has two continuations with exactly the same law through time N, one with atomic capacity one and finite infinite-horizon seed entropy, the other with atomic capacity zero and complete asymptotic forgetting.*

Proof. For the first continuation set $p_{N+k} = 2^{-k-3}$. Its binary entropies are summable. For the second set $p_{N+k} = 1/4$ for every $k \ge 1$. Apply Theorem 4.1. Future probabilities do not affect the common prefix law. □

Corollary 8.2. *Without restrictions on the unobserved tail, no statistic of a fixed finite-horizon record identifies infinite-horizon atomic capacity. This remains true even if the finite-horizon law is known exactly.*